

1. 孤立导体的电容

孤立导体，可证明 $V \propto q$ 定义孤立导体的电容：

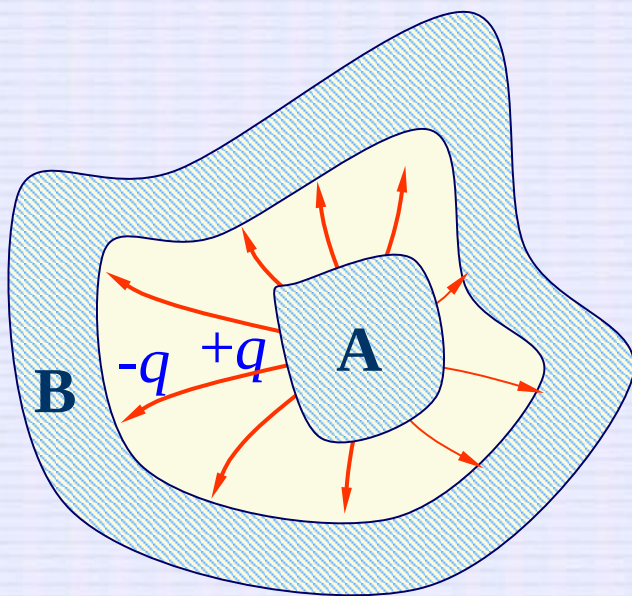
$$C = \frac{q}{V}$$

- C 决定于导体的形状、大小及周围的电介质，与导体带电荷无关。
- 单位 $1\text{ F} = 10^6 \mu\text{ F} = 10^{12} \text{ pF}$

2. 电容器的电容

2.1 电容器

空腔导体 B 与其腔内的导体 A 组成的导体系统，叫做**电容器**，A、B 为电容器的两极板，导体 A、B 相对两表面上所带的等值异号的电荷量为电容器所带的电量。



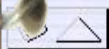
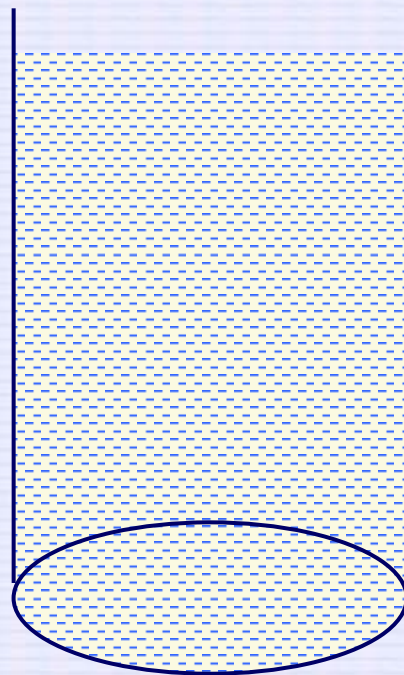
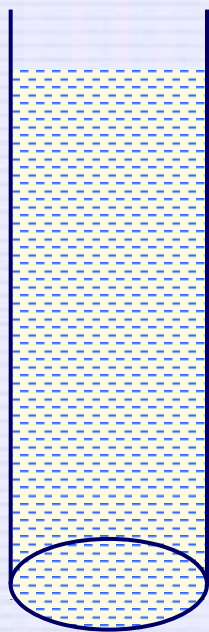
电容器

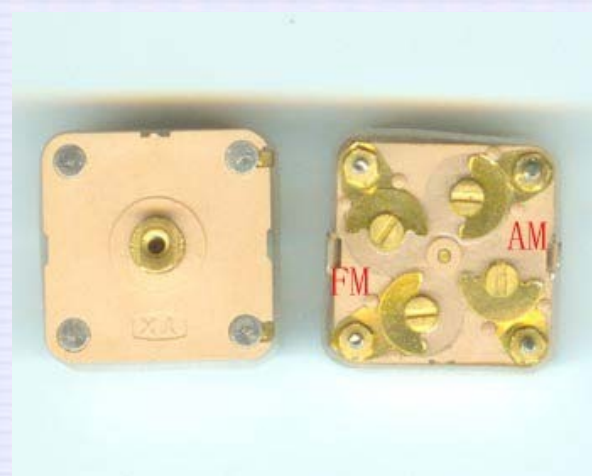
2.2 电容器的电容

~~电容器~~。定义电容器的电容：

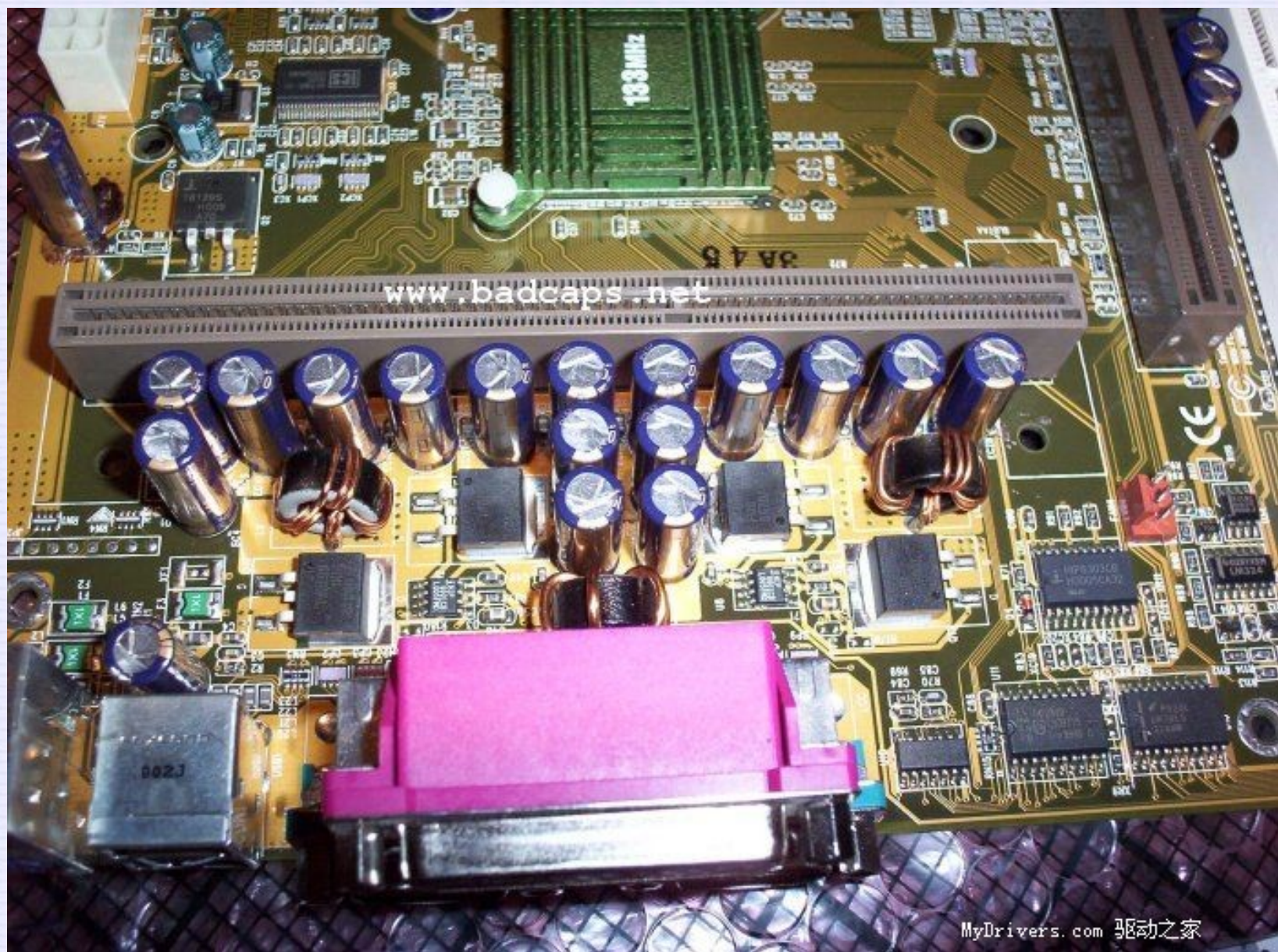
$$C = \frac{q}{V_{AB}}$$

- C 决定于两极板 A 、 B 的形状、大小， A 、 B 的相对位置以及 A 、 B 间的电介质，与电容器带电荷无关。





各种电容器



3. 电容器电容的计算

计算要点：

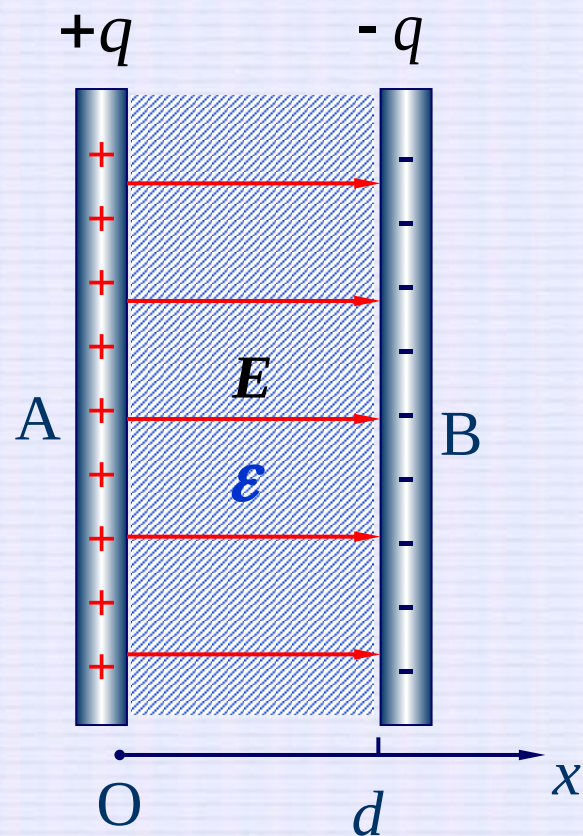
- 1) 设电容器带电 q , 求极板间场强分布 $E = E(r)$
- 2) 计算极板间的电势差: $V_{AB} = \int_A^B E \cdot dl$
- 3) 由电容器电容定义计算 电容: $C = \frac{q}{V_{AB}}$

3.1 平板电容器的电容

$$E = \frac{\sigma}{\varepsilon} = \frac{q}{\varepsilon S}$$

$$V_{AB} = \int_A^B \vec{E} \cdot d\vec{l} = Ed = \frac{qd}{\varepsilon S}$$

$$C = \frac{q}{V_{AB}} = \frac{\varepsilon S}{d}$$

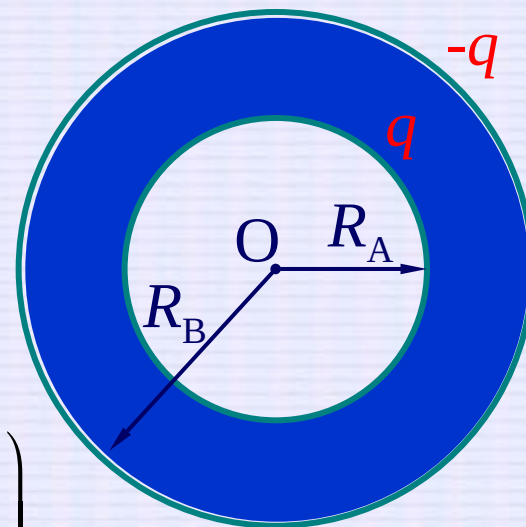


3.2 球形电容器的电容

$$E = \frac{q}{4\pi\epsilon r^2}$$

$$V_{AB} = \int_{R_A}^{R_B} \frac{Qdr}{4\pi\epsilon r^2} = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_A} - \frac{1}{R_B} \right)$$

$$C = \frac{q}{V_{AB}} = \frac{4\pi\epsilon R_A R_B}{R_B - R_A}$$



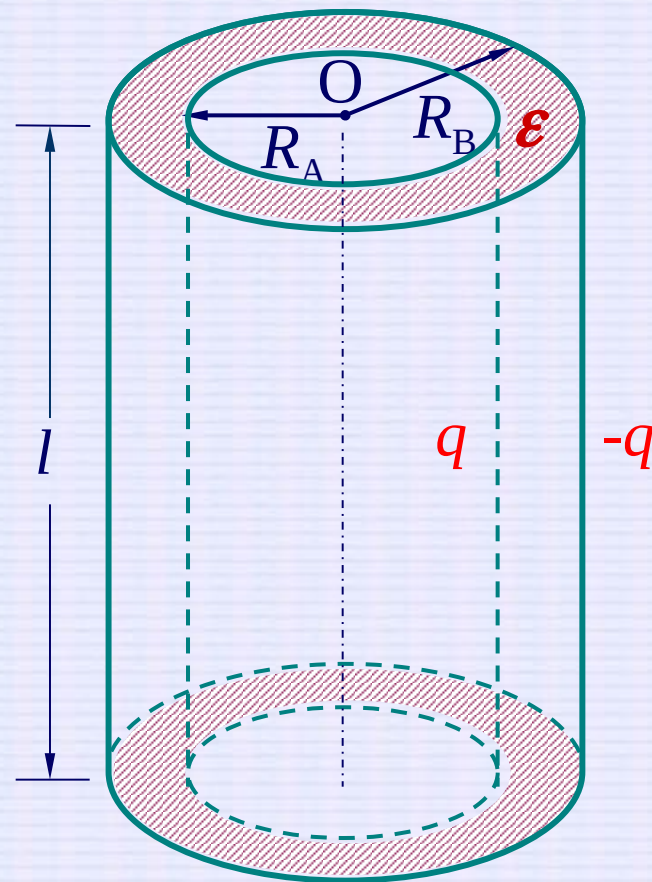
3.3 圆柱形电容器的电容

$$E = \frac{q}{2\pi\epsilon l r}$$

$$V_{AB} = \int_{R_A}^{R_B} E dr = \int_{R_A}^{R_B} \frac{q dr}{2\pi\epsilon l r}$$

$$= \frac{q}{2\pi\epsilon l} \ln \frac{R_B}{R_A}$$

$$C = \frac{q}{V_{AB}} = \frac{2\pi\epsilon l}{\ln R_B / R_A}$$

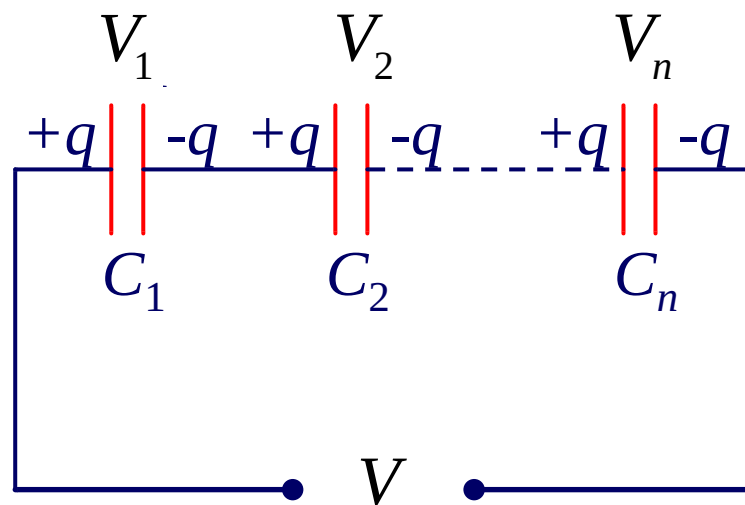


4. 电容器的串联和并联

4.1 电容器的串联

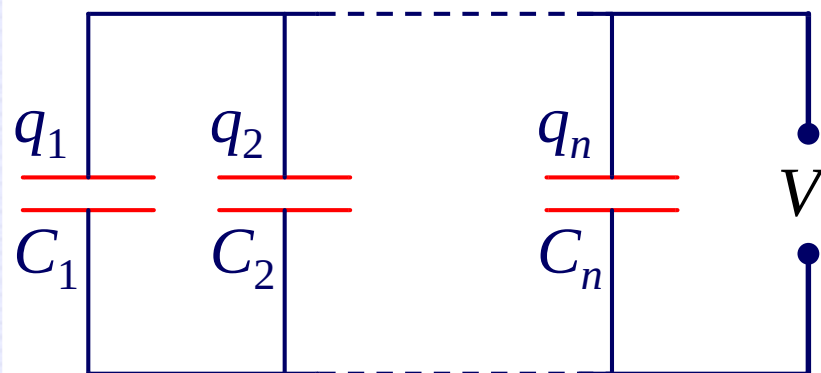
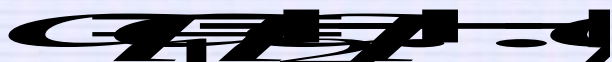


$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$



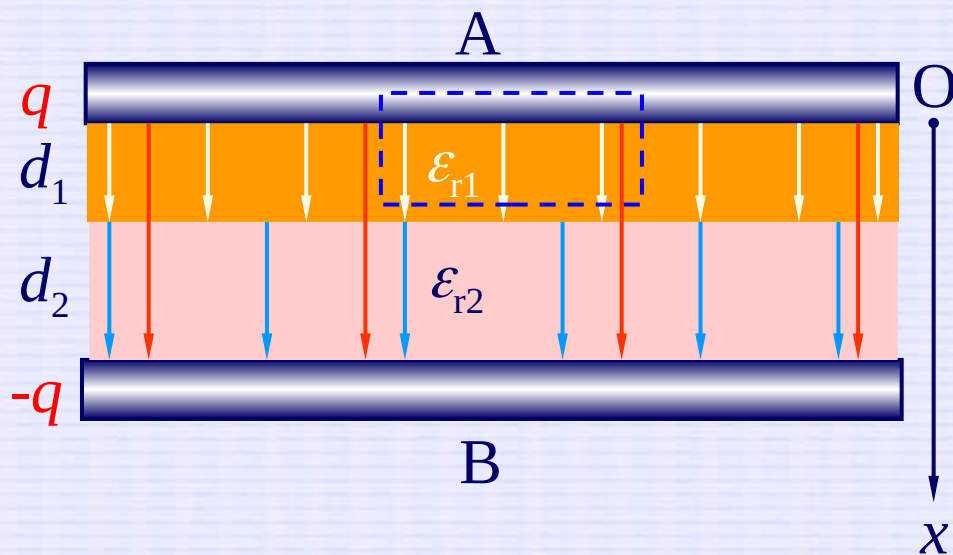
电容器的串联

4.2 电容器的并联



电容器的并联

例 8-5 一平行板电容器，中间有两层厚度分别为 d_1 和 d_2 的电介质，它们的相对介电常数分别为 ϵ_{r1} 和 ϵ_{r2} ，极板面积为 S ，求该电容器的电容。



解: $D = \sigma$

$$E = E(x) = \frac{D}{\varepsilon} = \begin{cases} \frac{\sigma}{\varepsilon_0 \varepsilon_{r1}} & (0 < x < d_1) \\ \frac{\sigma}{\varepsilon_0 \varepsilon_{r2}} & (d_1 < x < d_1 + d_2) \end{cases}$$

$$V_{AB} = \int_0^{d_1+d_2} E(x) dx = \int_0^{d_1} \frac{\sigma}{\varepsilon_0 \varepsilon_{r1}} dx + \int_{d_1}^{d_1+d_2} \frac{\sigma}{\varepsilon_0 \varepsilon_{r2}} dx$$

$$= \frac{\sigma}{\varepsilon_0} \left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} \right) = \frac{q}{\varepsilon_0 S} \left(\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} \right)$$

$$C = \frac{q}{V_{AB}} = \frac{\varepsilon_0 S}{d_1 / \varepsilon_{r1} + d_2 / \varepsilon_{r2}}$$

作业：

6-13 、 6-16

